

Zemar Hakim

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Education

Queen's University

PhD, Economics

Candidate (Expected 2027)

Toronto Metropolitan University

MA, International Economics and Finance

Wilfrid Laurier University

BA, Honours Economics

Research

Applied Econometrics, Causal Inference, Environmental and Climate Economics, Development Economics

Proximity to the Plague: Measuring Locust Exposure and Child Health. Job Market Paper. [\[PDF\]](#)

Abstract: I investigate the early childhood health outcomes of the Kenyan 2020–2021 desert locust invasion. To account for the multi-harvest setting, I build a treatment that conditions on the insect's biological capacity to damage vegetation, the growth stage of the local crop, and subsequent vegetal decline, matching georeferenced locust observations to the 2022 Kenya Demographic and Health Survey. Children exposed in utero within 30 km of a verified damage event carry weight-for-height z-scores 0.185 standard deviations below the control group, and are 26 percent more likely to be clinically wasted. Effects are strongest when children are exposed in the second and third trimester. Riverine livelihood zones, where households depend heavily on their own harvest and have the least market cushion lose an additional 0.25 standard deviations on weight-for-height beyond the average cropping-zone effect.

Wildfire Smoke, Air Quality, and the Limits of Short-Run Labour-Market Impacts in Canadian Cities. With Christopher S. Cotton. Submitted to *Nature Climate Change*, August 2026. [\[PDF\]](#)

Abstract: Wildfire-smoke-driven increases in fine particulate matter raise hours worked and employment in Canada's nine largest metropolitan areas. This response is consistent with productivity impairment requiring longer hours to meet output targets and with degraded outdoor conditions reducing the value of non-work time. Both mechanisms imply welfare losses that standard labour-market statistics record as gains, suggesting output-based damage assessments understate the costs of smoke in urban economies.

When It Never Rains: Index Insurance and the Resilience of Relational Contracts During Drought. Working paper, April 2026. [\[PDF\]](#)

Abstract: A central concern in the microinsurance literature is whether formal insurance contracts crowd out informal risk-sharing arrangements that underpin livelihoods in developing economies. Using six rounds of panel data from pastoralist households in Marsabit, northern Kenya, we estimate the causal effect of Index-Based Livestock Insurance (IBLI) on informal livestock lending during drought. We instrument IBLI uptake with randomly distributed price discount coupons and exploit exogenous variation in drought severity measured by the Standardized Precipitation-Evapotranspiration Index (SPEI). Our two-stage least squares estimates, corrected for small-cluster bias with 16 sub-location clusters, show that IBLI coverage significantly increases the probability of borrowing livestock during drought—a crowding-in effect, not crowding out. A one-standard-deviation worsening of drought conditions raises the probability of livestock borrowing by roughly 39 percentage points more for insured than uninsured households. Placebo tests on livestock gifts and outward loans show no significant effects, supporting the exclusion restriction. Generalized random forest estimates reveal that treatment effect heterogeneity is driven primarily by drought severity and livestock wealth. These findings suggest that formal index insurance complements, rather than substitutes for, informal reciprocal lending networks, providing a new channel through which microinsurance improves household resilience.

Born into Drought: Early-Life Exposure and Child Health in Angola. Work in progress.

Policy Papers and Other Contributions

Patent Monetization Promotion and SSO Participation. White paper. With Charles River Associates, 2026.

G7 Sanctions Matching Report. Policy report. With Global Affairs Canada. Presented during Canada's G7 Presidency, 2025.

Conferences

Applied Economics Conference, Institute of Economic Sciences, Belgrade (2026): *Wildfire Smoke, Air Quality, and the Limits of Short-Run Labour-Market Impacts in Canadian Cities*

Canadian Economics Association Annual Meetings, Simon Fraser University, Vancouver (2026): *Born into Drought: Early-Life Exposure and Child Health in Angola*

Canadian Economics Association Annual Meetings, Université du Québec à Montréal (2025): *Proximity to the Plague: Measuring Locust Exposure and Child Health*

Canadian Development Economics Study Group, Concordia University, Montréal (2025): *Proximity to the Plague: Measuring Locust Exposure and Child Health*

Awards

Mac Urquhart PhD Award for best second-year paper in Economics (\$2,500)	2026
Economics Department Award (\$10,000 x 4)	2022-2026
E. Gay Mitchell Graduate Fellowship in Finance (\$8,000)	2024
Ontario Graduate Scholarship (\$15,000 x 2)	2020, 2022
Dean's List (\$1,500)	2018
Wilfrid Laurier Entrance Scholarship (\$15,000)	2015

Professional Experience

Senior Associate (Summer Intern) Toronto, Ontario
Charles River Associates 2026

- White Paper: Patent Monetization Promotion and SSO Participation.
- Conducted quantitative and economic analysis for 5 litigation and merger engagements.
- Built automated codebook to support replication.
- Built and maintained econometric models in R and Stata to evaluate damages and market dynamics.
- Conducted industry research and translated findings into clear reports and presentations for clients.

Senior Research Analyst (Consultant) Ottawa, Ontario
Global Affairs Canada 2022-2026

- Policy Paper: G7 Sanctions Gap Analysis
- Designed automated data pipelines in R, Python, and SQL with over 500 million records from 4 federal departments.
- Developed causal and predictive models to quantify the impact of sanctions and conflicts, forecasting trade flows worth more than \$2 trillion annually and identifying anomalous transaction patterns across G7 economies.
- Managed 2 direct reports and supported over 20 ministerial briefings.
- Engineered architecture integrating over 25,000 records across allied jurisdictions, and built interactive dashboards for 3 federal departments.

Economics Expert (Contract) 2025-2026
Mercor

- Designed AI training data for large language models across 4 projects.
- Diagnosed model failure modes in logic, numerical reasoning and code generation.

Research Assistant

2020-2026

- Philippe Kabore: Data cleaning and analysis
- Cherie Metcalf: Data visualization and basic analysis
- Hyunju Lee: Literature Review

Teaching Assistant

2020-2026

- ECON 290: Energy Economics
- PPEC 200: Introduction to the Study of Politics, Philosophy and Economics
- ECON 110: Head TA, Principles of Economics
- ECON 111: Head TA, Introductory Microeconomics
- ECON 112: Head TA, Introductory Macroeconomics
- ECON 430: Issues in Economic History
- ECON 831: Issues in North American Economic History
- ECN 606: International Monetary Economics

President and CEO

Waterloo, Ontario

Wilfrid Laurier University Students' Union

2019 – 2020

- Elected to represent over 17,000 students, managing an \$11 million budget and a team of 5 direct reports to deliver operations, services, and advocacy across institutional, municipal, provincial, and federal levels of government.
- Created and led the university's first institutional committee on student basic needs, addressing food insecurity and housing, and sustained service continuity through emerging crises including the early stages of the COVID-19 pandemic.

Technical Skills

Programming Languages: R, Python, SQL, Stata, MATLAB, ArcGIS, QGIS, PostGIS

Cloud and Collaboration: GitHub, Microsoft Office (incl. Power BI), Google Suite (incl. GCP), AWS (S3, EC2), Jupyter, L^AT_EX, Google Earth Engine, Databricks

Languages: English, French, Urdu

Volunteering and Community Activity

Shepherds of Good Hope - Ottawa Homeless Service
Mississauga Food Bank

References

References available upon request